AUG
12

FA1_Creating Research Questions_ATL Research Skills
Ishita Bhattacharjee
Formative Formative 21/21 Students Wednesday at 10:30 AM
IB MYP IB MYP Design - Ms Ishitha (MYP4)

Task Details

Based on the design problem you are currently developing, **construct** 10 research questions to establish a robust research plan for your solution.

For each question:

- Identify which **type of question** it is, drawing on the 8 types introduced in class (Factual, Comparative, Causal, Exploratory, Evaluative, User-centred, Predictive, Technical/Feasibility). Use a mix — do not rely on just one or two types.
- State whether the data will come from **primary research** (you collect it yourself — e.g. survey, interview, observation, testing) or **secondary research** (existing sources — e.g. articles, standards, product reviews, websites).
- Name the **specific source or method** you would use to answer it (e.g. "survey of 10 classmates," "manufacturer's spec sheet," "interview with a PE teacher").

Success criteria:

- All 10 questions relate directly to your own design problem.
- Questions cannot be answered with a 30-second Google search alone.
- At least one question would help you justify a design decision later in Criterion B.
- Primary and secondary sources are both represented, and each is specific (not just "the internet").

Record your work on the Research Question Practice sheet.

Research_Question_Practice_Worksheet.docx10.61 KB

Attaching the detailed Task specific clarification here.

Band	Research Plan Requirements	Indicators / Evidence of Achievement
7–8	Constructs a **detailed and independent research plan** using a table, flowchart, or mind map. Identifies research areas, constructs relevant research questions, and clearly prioritizes the primary and secondary research needed to develop a solution.	• Formulates **10 highly focused and insightful research questions** directly addressing the core design problem.• Demonstrates mastery through a **well-balanced variety across all 8 research question types**.• Accurately identifies and names **highly specific, appropriate primary and secondary research sources/methods** tailored to each inquiry.• Research questions directly support and justify **specific design decisions, material selections, technical requirements, and user constraints**.
5–6	Constructs a **detailed research plan with some guidance** using a table, flowchart, or mind map. Identifies research areas, constructs	• Develops **10 clear and relevant research questions** directly linked to the design problem and requiring deeper inquiry.• Uses a **diverse mix of approximately 5–6 research question types** across the required categories.• Correctly identifies and names **specific primary and secondary sources/**

Band	Research Plan Requirements	Indicators / Evidence of Achievement
	relevant research questions, and prioritizes the primary and secondary research needed to develop a solution.	**methods**, such as a survey of a target group or a manufacturer specification sheet • At least one or more questions clearly help justify potential **design decisions**.
3–4	Outlines a research plan with some guidance. Identifies the **primary and secondary research** needed to develop a solution and provides research questions, research type, and basic justification.	• Formulates **at least 5 relevant research questions**, with most of the expected 10 questions connected to the design problem.• Uses around **3–4 different research question types** across the required categories.• Identifies whether research is **primary or secondary**, although source or method descriptions may be general, such as “a survey” or “articles.”• Provides a basic foundation for research but shows **limited justification or prioritization** of how findings will influence future design decisions.
1–2	Produces a **limited or incomplete research plan** with significant guidance. Research needs, questions, and the distinction between primary and secondary research may be unclear.	• Drafts **fewer than 10 research questions**, or questions lack sufficient depth and may be answerable through a quick search.• Relies heavily on only **1–2 research question types**.• Identifies research sources vaguely, for example, simply stating **“the internet.”**• Shows minimal distinction between **primary and secondary research**.• Research questions provide little connection to or justification for future design decisions.
0	Does not reach a standard described by any of the descriptors.	• No research questions are submitted, or the questions provided do not relate to the design problem.

Assessment Criteria

A: Inquiring and analysing

Achievement Level	Descriptor
0	The student has not submitted any research questions, or the questions provided do not relate to the design problem.
1 - 2	The student: • Drafts fewer than 10 research questions or questions that lack depth (can be answered with a 30-second Google search). • Relies heavily on only 1 or 2 question types. • Identifies research sources vaguely (e.g., just listing "the internet"). • Shows minimal distinction between primary and secondary research.
3 - 4	The student: • Formulates most of the 10 research questions relevant to the design problem. • Uses 3–4 different question types across the 8 required categories. • Identifies whether data is primary or secondary, but source/method names are slightly general (e.g., "a survey" or "articles"). • Questions provide a basic foundation for research but limited justification for future design decisions (Criterion B).

Achievement Level	Descriptor
5 - 6	The student: • Develops 10 clear, relevant research questions directly tied to the design problem that require deeper inquiry. • Develops 10 clear, relevant research questions directly tied to the design problem that require deeper inquiry. • Uses a diverse mix (5–6 types) of the 8 question categories (Factual, Comparative, Causal, Exploratory, Evaluative, User-centred, Predictive, Technical/Feasibility). • Correctly categorizes and names specific primary and secondary sources/methods (e.g., "survey of 10 classmates," "manufacturer spec sheet"). • At least one question clearly helps justify potential design decisions.
7 - 8	The student: • Formulates 10 highly focused, insightful research questions directly addressing the core design problem. • Demonstrates mastery by using a well-balanced variety across the 8 question types. • Accurately classifies and names highly specific, well-chosen primary and secondary sources/methods tailored to each inquiry. • Questions directly lay the groundwork to justify specific design choices, material selections, or user constraints.